

A host of varieties of end-effectors are available for diverse applications of robotics. Some robots may have spray paint dispensers, measurement tools, solder guns, glue applicators as end-effectors, others may have welding torches, cutters, CNC wires, machine tools.

The end-effector being a complex being of equipment often requires its own control circuitry, and this may be independent of the rest of the robot, or integrated. The end effector pictured above, with only one degree of freedom, illustrates the complexity that an end-effector may have to accommodate to perform even simple functions. Adding to the complexity, such as the design of a fully functional anthropomorphic human hand (which possesses 14 knuckle joints, and an infinite number of possible ways to grip almost any given object) requires a great deal of optimisation, intelligent programming and intuition for geometry and mechanics.

How it obtains information - sensors

It is fairly common to find robotics applications where a huge amount of power or manoeuvrability is provided by the machinery in a way that needs to be skilfully manipulated by a human being. Consider a digger (also known as an excavator), which has powerful, accurately controlled robotic arms with a huge spade attached at the end. This potent machine still required the expertise of a human being behind the wheel, directing it exactly in the way it needs to move.



A remote sensing protective glove that integrates tactile, orientation and electrical sensors.

Several robotics applications would become pretty redundant if all robots

required the constant supervision of a human being. Consider the post-sorting robots used by the US Postal Service - imagine if a human being had to constantly check that the robotics were not randomly chucking envelopes into different bins!

That's where sensors come in. A good robot designer is able to identify exactly what kinds of information will be most useful to a robot in performing its task. Accordingly sensors of light, heat, proximity, sound, and even chemicals have been developed.